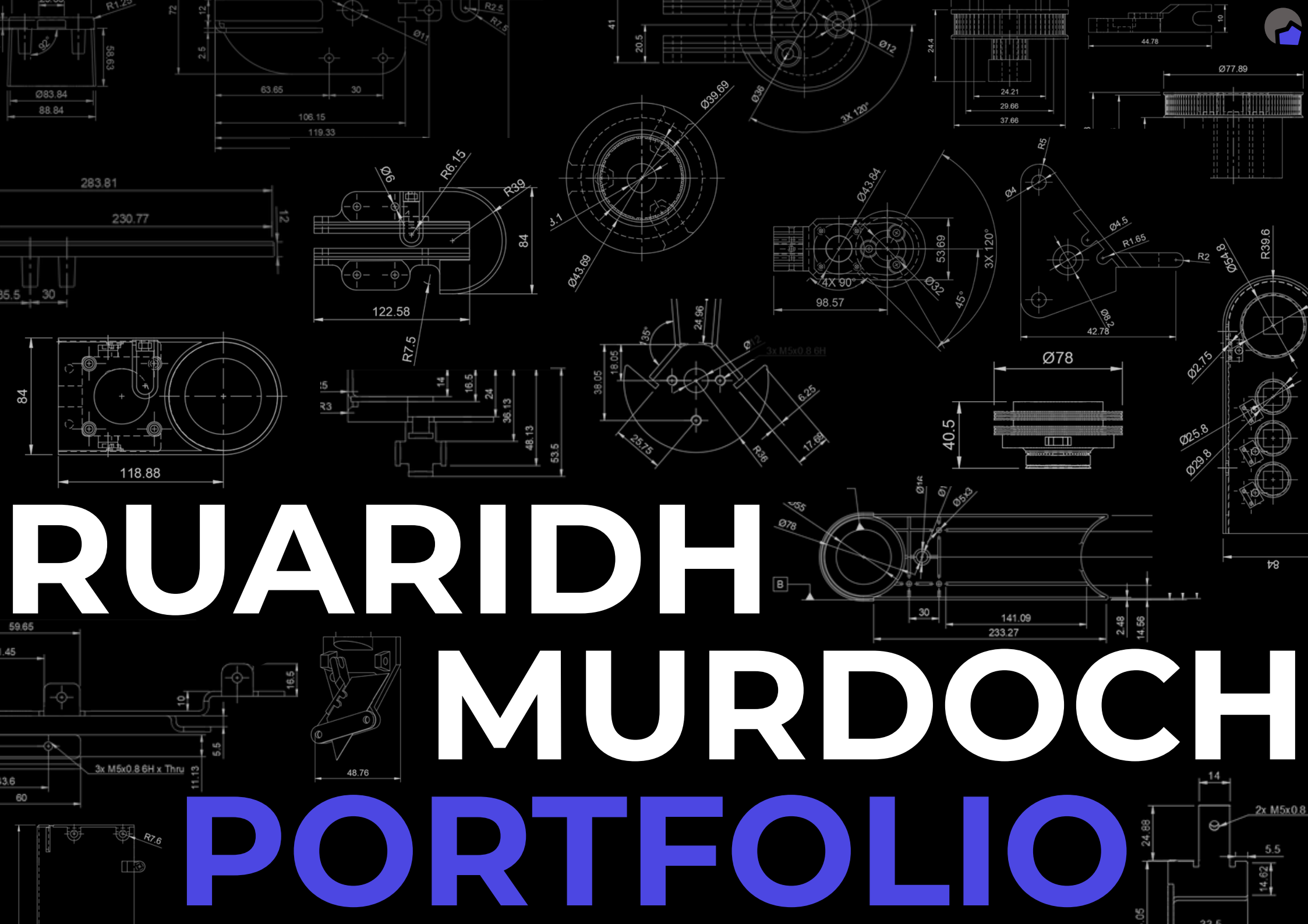




RUARIDH MURDOCH PORTFOLIO

About Me

I'm a third-year Design Engineering student at Imperial College London, **ranked first in my cohort** and awarded the **Head of School Prize**. I like getting to the bottom of a problem by **talking to the people who have it**, then working across mechanical design, electronics and software to fix it. My main strength is CAD.

I'm most interested in **physical products people use every day**, where details like how a mechanism feels or how long it lasts decide whether people enjoy using them. I'm looking for a placement where I can learn fast.

Outside university, I'm a NIBAS- and NICAS-certified lead climber and former Glasgow Youth Squad competitor. I've completed the Lakeland Mountain Marathon, hold an RYA Day Skipper licence and make games in my spare time.

Full reports, videos, live demos and more projects are on my website:
ruaridhmurdoch.com





STAN

A Teach-by-Touch Robotic Arm

Programming a robot usually means writing code. STAN removes that barrier by making you **teach it by hand**. The brief was a mechatronic system on a **tight budget**, using simple electronics and few components.

STAN lets anyone create movement sequences by physically positioning its joints. Motions are recorded, stored and replayed with **colour-coded cartridges** on demand, turning programming into a tangible and playful interaction **without requiring any coding knowledge**.

Joints built from 5V DC motors, gear reductions and potentiometers rather than integrated servos. All CAD, code and structure **produced in-house**, mostly 3D printed.



My Contributions

Joint Design & CAD
Embedded Software

Rapid Prototyping
Mechanical Design

Group of 2

2



STAN

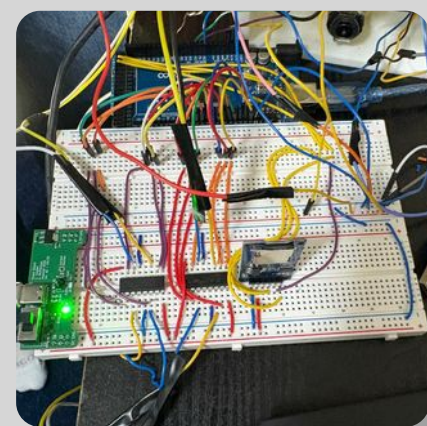
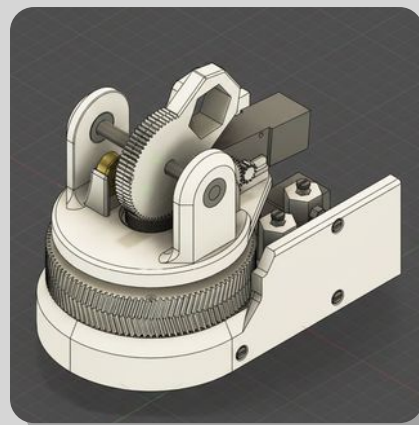
Development Process

The joint mechanism went through **multiple full redesigns** before settling on the final geartrain and bearing layout. Early versions used different gear ratios and motor mounts, tested and rejected based on backlash, torque, and how cleanly they could be 3D printed and assembled.

Each sub-assembly was iterated individually; **print, test, fail, redesign**, before being combined into the working arm.

As the system developed, the arm geometry, transmission ratios and internal layout were refined to improve **movement, strength and reliability**.

The cartridge interface and electronics were then integrated into the final assembly, creating a resourceful system built largely from **custom 3D-printed components**.





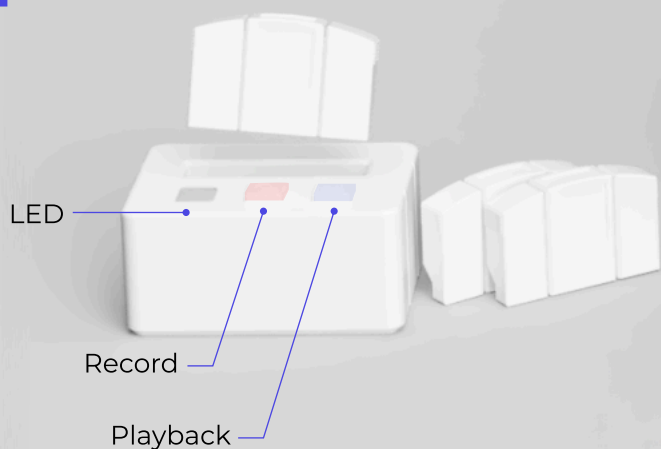
STAN

Final Design

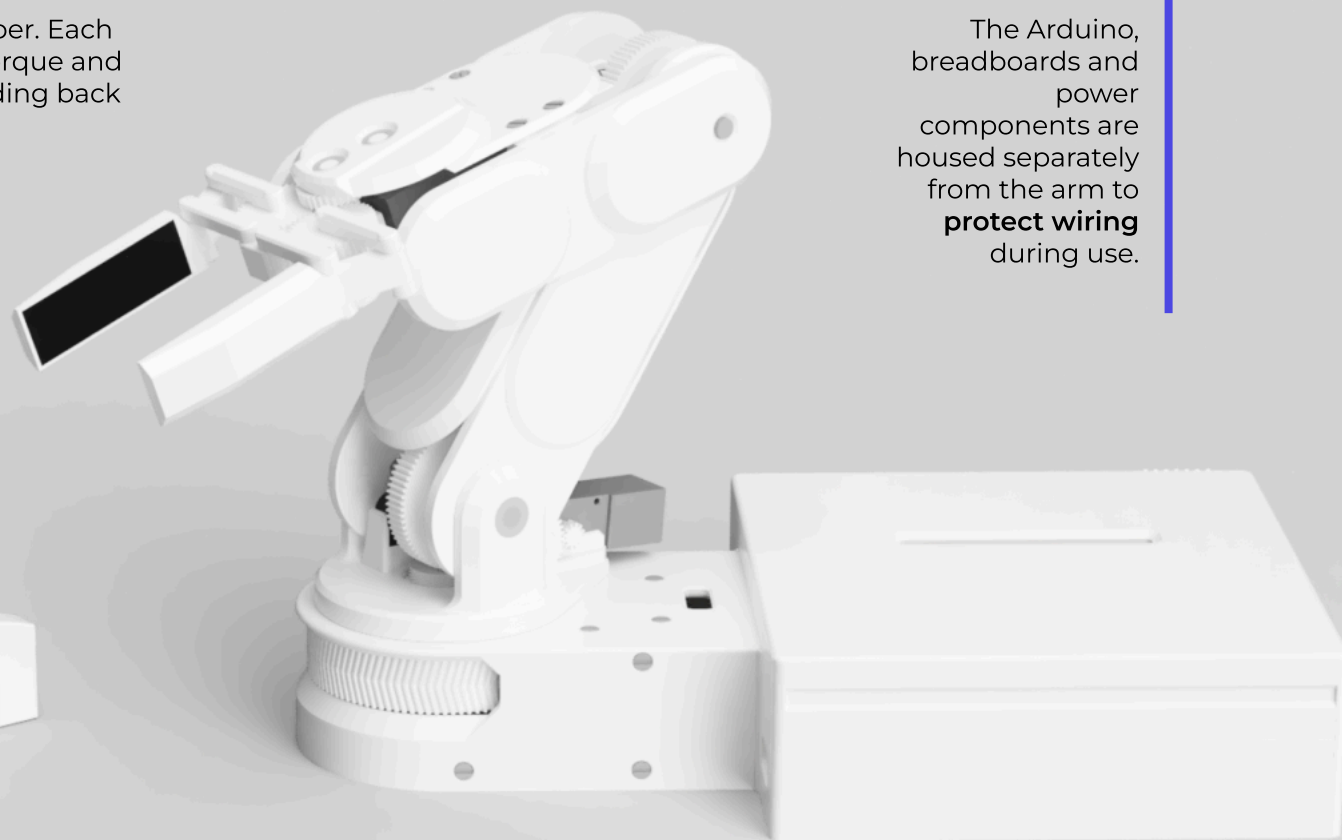
Four joints: swivel, shoulder, elbow and gripper. Each runs through a gear reduction to increase torque and control resolution, with **potentiometers** feeding back exact position as the arm is moved by hand.

Each recording stores up to 800 frames, providing around **20 seconds of motion**. Record captures joint angles in real time, while Playback reproduces the motion at its original speed.

The Arduino, breadboards and power components are housed separately from the arm to **protect wiring** during use.



Each cartridge has a colour patch read by an RGB sensor, linking it to a sequence. Swapping them **instantly loads the matching recording**.





SwivelCut

A Desktop Cardboard Cutting Tool

Design & Technology students use cardboard for early-stage prototyping, yet accurate cutting is often **slow, inconsistent and unsafe**. Craft knives require skill, produce variable results and can **discourage students from iterating**.

SwivelCut is a safe desktop tool that lets students **trace a design** before reproducing it with **consistent accuracy**. Interchangeable heads and integrated safety features make cardboard prototyping faster, more accessible and less reliant on traditional cutting methods.



My Contributions

Concept Development

User Testing

DFMA

Visual Design

Mechanical Prototyping

Videography

Group of 4

5



SwivelCut

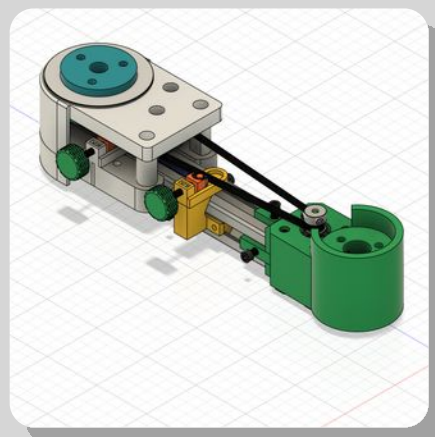
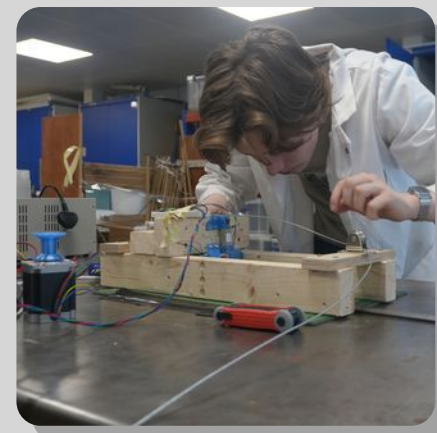
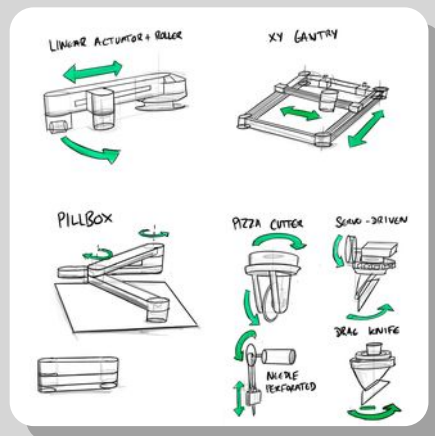
Research & Design Process

Early concepts were developed through **morphological analysis**, comparing different mechanisms before selecting a 2DOF flat arm with interchangeable heads. **Foam and card prototypes** were used to test joint movement, workspace and reach before progressing into CAD.

Custom test rigs then measured the **piercing and drag forces** required to cut cardboard, helping define the motor torque, gear reduction and structural requirements.

The joints were **belt-driven** rather than directly geared, allowing tensioners to absorb build tolerances without relying on tightly machined fits. The prototype was largely 3D printed before being redesigned for manufacture using sheet metal and injection moulding, with **DFM and DFA** principles applied throughout.

A DT teacher's **visually impaired student** is why every button carries a **raised symbol you can read by touch**. Students and teachers shaped clamping, controls and accessibility throughout.





SwivelCut

Final Product

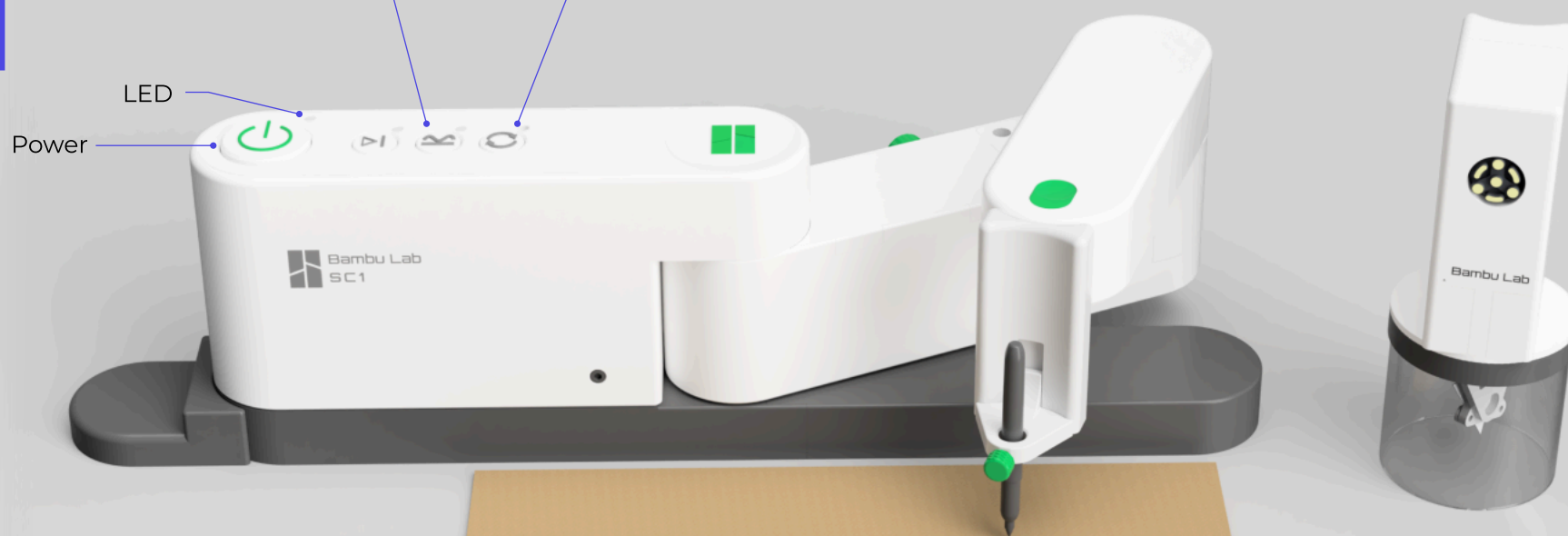
Trace a path by hand with the tracing head, then swap to the cutting head to follow that same path and cut it. The base clamps to a workbench from the end, and battery, motors and electronics are all housed inside.

Repeat: re-runs the last stored path for batch cutting

Stabilisation: smooths wobble in the hand-traced path before cutting

Both arms **fold flat** for compact storage and give **A3 cutting coverage**. Aluminium-reinforced structure with thrust bearings supporting each joint.

Cutting head features a spring-mounted blade on a linear actuator, enclosed by a **removable safety tube**. Rotating wheel shows when the blade is cutting.



18 students used it independently **after one demo**. Their teacher would consider several for his classroom.

White and dark grey casing, referencing the Bambu A1.



Rhythm Rush

Motion-Tracked Social Play for Queues

Concert queues are full of strangers who already share an interest, yet most people spend the wait **on their phones**. Our research found people **wanted to connect** but worried about intruding or looking like “a weirdo”.

Rhythm Rush is a **motion-tracked dance game** built into the queue. Up to four players step onto marked spots and appear as characters on a big screen, dancing along with an on-screen guide. Everyone else can watch, join in or carry on past **without losing their place**.



My Contributions

User Research

User Testing

Co-design

Social Experiments

Body Tracking

Concept Selection

Group of 5

8



Rhythm Rush

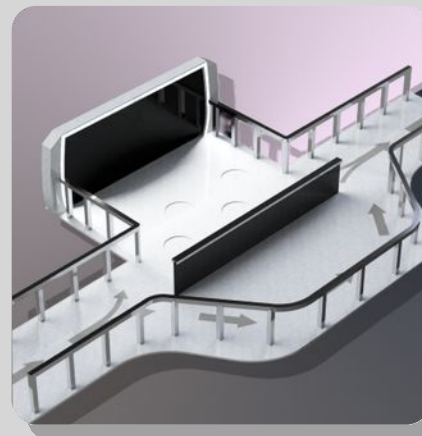
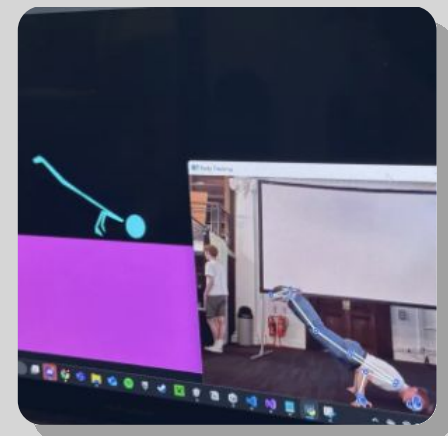
Research & Design Process

We started at Outernet, an immersive space by Tottenham Court Road built to get the public interacting. Visitors stayed **41 seconds** on average, nearly half were on their phones and **nobody spoke to anyone new**. Street interviews with 21 people told us why. In an **individualistic society**, most assumed strangers were busy and didn't want to be disturbed, and worried about coming across as strange.

Outside Imperial's library we offered passers-by a high five or a quick chat, then swapped the chat for a **two-player game**. **Interactions doubled**, and people started talking without being asked.

Follow-up interviews and a survey pointed us to **concert queues**. We took three concepts to users, using PrEmo emotion cards to see how each would feel to come across alone and with friends, and a Harris profile to score learnability and fit with the pace of a queue. **Rhythm Rush came out on top**.

I built the body tracking in Python with **MediaPipe**, testing it on people of different builds doing dramatic poses and jumps. Splitting the camera view gave each player **their own zone**, so scores never got mixed up.



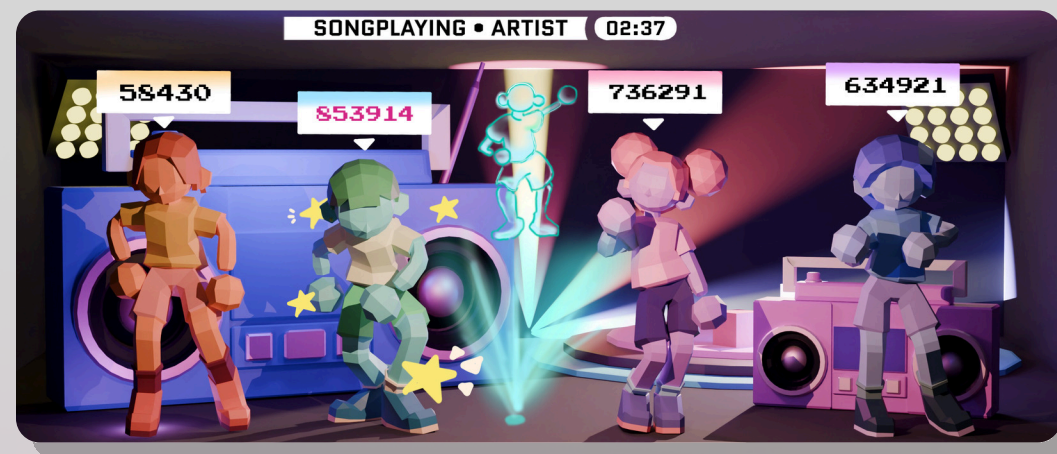


Rhythm Rush

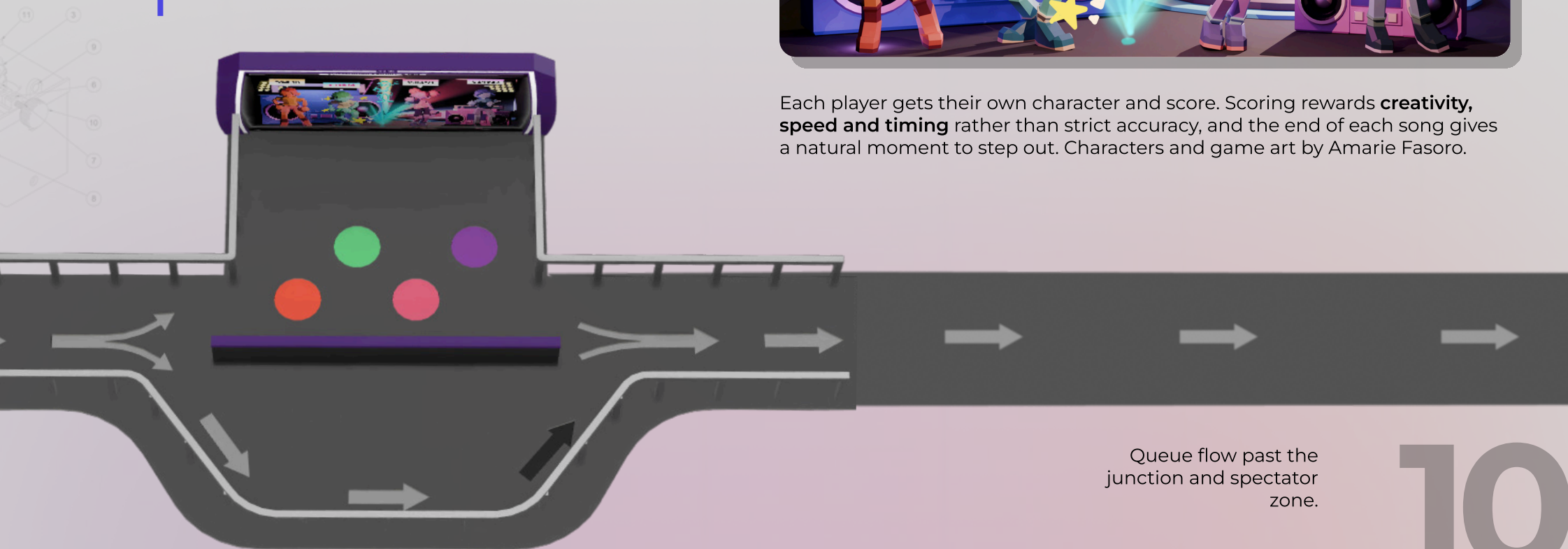
Final Design

In a stakeholder role-play and a simulated queue test, people's biggest worry was **losing their place**. The final layout adds a **junction** where each person chooses to play, watch from the spectator zone or carry on queueing.

The four player spots are staggered so everyone has room to move and can still see the screen, sized with a **full-scale mock-up**.



Each player gets their own character and score. Scoring rewards **creativity, speed and timing** rather than strict accuracy, and the end of each song gives a natural moment to step out. Characters and game art by Amarie Fasoro.



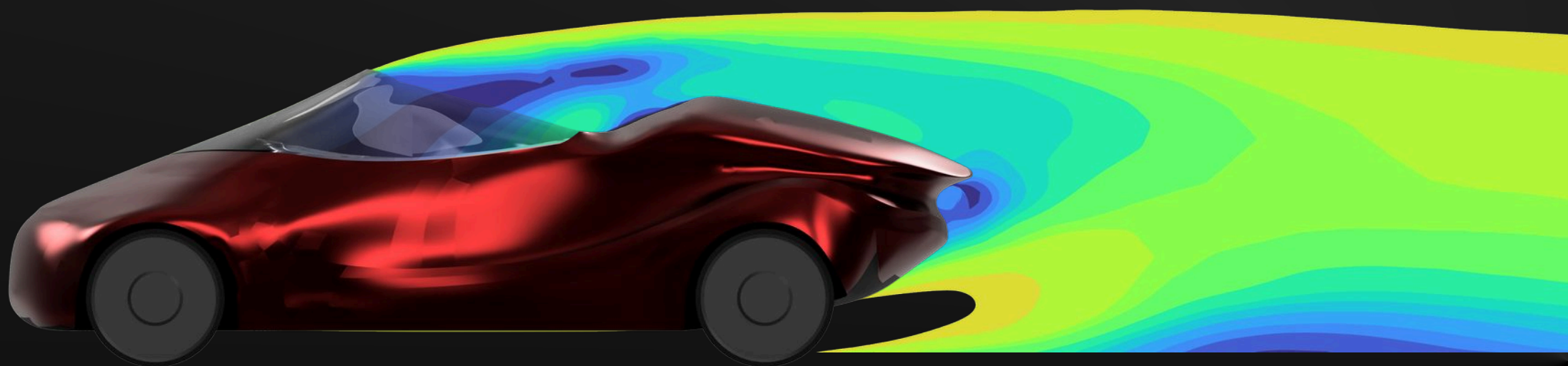


Aero. Cabriolet

Computational Fluid Dynamics

Drop the roof on a four-seat cabriolet and air tumbles through the cabin and out of the back. For the people inside that means **buffeting and wind noise**, and for the car a bigger wake and **more drag**.

I designed an electric four-seat cabriolet for **everyday roof-down driving**, shaped to send air **around the cabin rather than through it**. The body was modelled in Fusion 360 and analysed with CFD in Ansys.



Velocity contour at 30.5 m/s (68 mph). The longer rear body and side channels let the air **slow down and reattach** before it separates, keeping the wake behind the car narrow.



Aero. Cabriolet

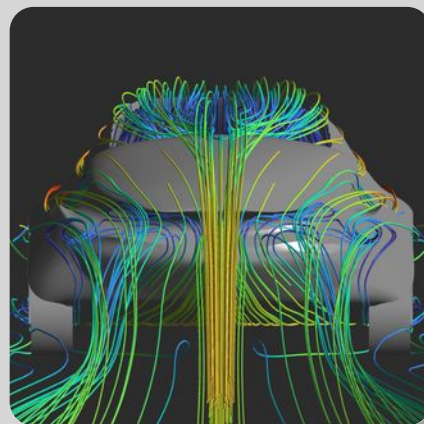
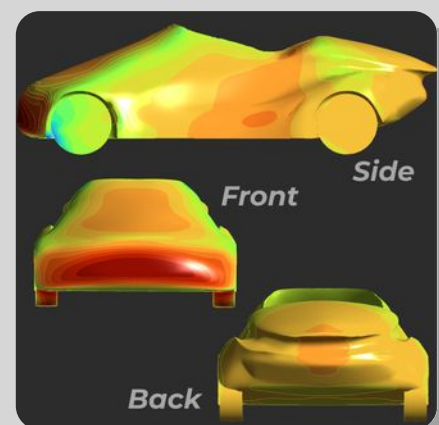
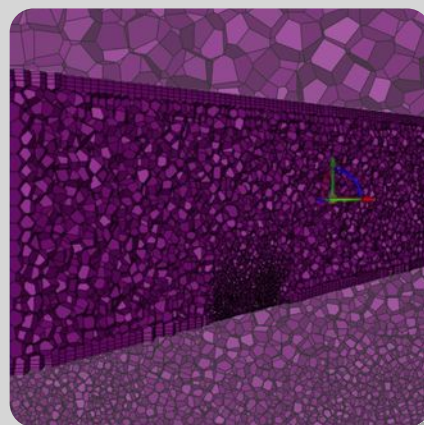
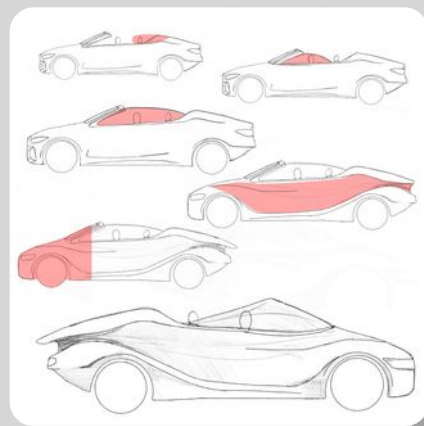
Design Process

I started from what **roof-down drivers care about**, low wind noise and good efficiency at motorway speeds. Benchmarking the BMW 4 Series Convertible, Porsche 911 Speedster and others showed most cabriolets leave the air exiting the cabin **unmanaged**.

Sketches developed the rear first: a **raised hump** behind the seats gives cabin air something to reattach to, followed by a longer, tapered tail.

Going electric removed the need for cooling inlets, so the nose could be clean and the undertray **fully flat**. The cabin was narrowed and partly enclosed at the sides so more air passes around it, while **side channels** feed faster air into the wake.

Running the model through Ansys Fluent showed high pressure at the nose and air entering from the sides. Refining both cut the **drag coefficient by 12%** and **drag force by 14%**.





Aero. Cabriolet

Wind Tunnel Testing

The car was 3D printed at 1:23 scale in orange, so white strings taped to the body would show up clearly. As a group of four we tested it in Imperial's **T1 wind tunnel** at 10, 20 and 31.5 m/s.

At 31.5 m/s the measured drag coefficient **closely matched the CFD**, which gave us confidence in the simulation. Lift didn't match, which aero PhD students helped us trace to **gaps around the mounting plate**.

Strings at the rear of the cabin flapped the most. Compared with the CFD, this showed air still curling in around the windshield and **recirculating inside**.

The side channels and rear underside worked as intended. The next step would be a **flatter, longer nose** to shrink the high-pressure zone at the front.





Modular Economy Seat

Sustainable Design For Long-Haul Flights

Long-haul passengers sit in the same economy seat for **6-12 hours**, yet the seat is built around an **average male body**. Shorter, taller and broader people end up with poor sleep, neck strain and pressure points.

In a group of four we surveyed **122 passengers**, mapped everyone who deals with the seat and **took a real seat row apart**. I then designed a modular economy seat that fits more bodies and keeps its cushions in service for longer.



We stripped a 1999 Safran Weber row down to every part. Glued foam and velcro meant the cushions couldn't be separated, so they end up as **mixed waste**.



Cushions and covers are replaced roughly **every two years** while the aluminium frame lasts far longer. That made the soft parts the point where **comfort, cleaning and waste** all meet.





Economy Seat

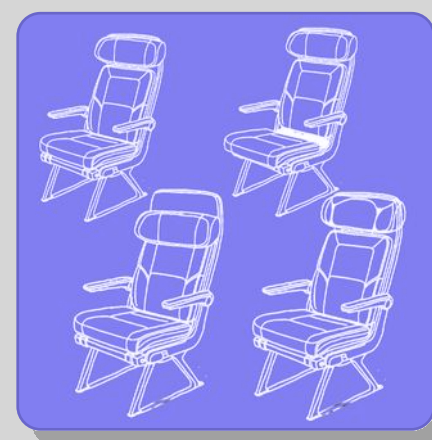
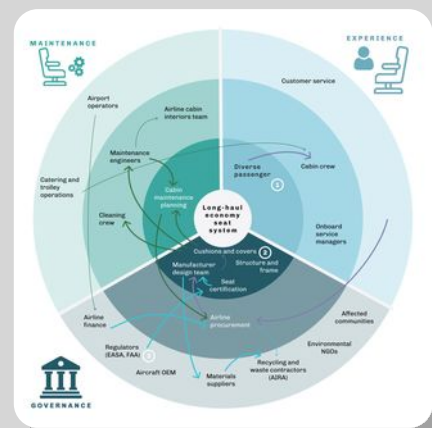
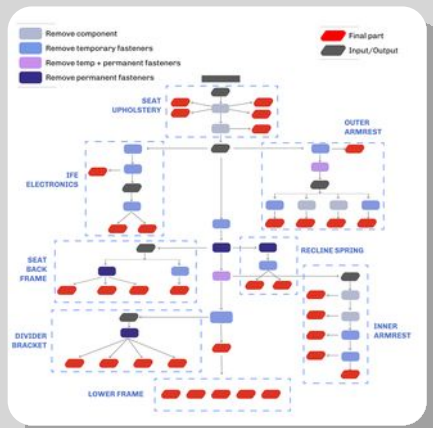
Research & Concept

Adjustable headrests scored just **2.25 out of 5** in our survey, and **41.8%** of passengers bring their own pillow. Two personas came from this: Petite Priya, whose feet dangle and headrest sits too high, and Broad Ben, who won't recline for fear of invading others' space.

Stakeholder mapping showed passenger complaints rarely reach the people who design the cushions, while cleaners and maintenance crews **see the wear first**.

Benchmarking car seats and office chairs pointed to a few **simple adjustments rather than a new frame**: a sliding seat pan for different leg lengths, a height-adjustable headrest with **fold-out wings** that mark out personal space, and a softer front edge to ease pressure under the thighs.

Spacer fabric and zoned foam reduce heat and sweat build-up, so passengers **fidget less** and the cushions wear more evenly.



Systems & Lifecycle Design

